Hsin Jee Sum

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EDUCATION

University of Leeds **Leeds, UK**
MEng, BSc Computer Science with Artificial Intelligence Sep 2022 – Jul 2026

- Average Grade: 75% (First Class Honours)
- Relevant modules: OOP in Java (81), Algorithms & Data Structures (78), Operating Systems (71), Advanced SE (80)

WORK EXPERIENCE

Leeds Gryphon Racing, Formula Student **Leeds, UK**
Visions Team Lead; AI R&D Division Sep 2025 – Jun 2026

- Led 10+ developers in sprints, prioritizing features weekly to deliver an autonomy stack, 24th of 59 at FS-UK 2026
- Optimized an inherited YOLOv8 model to a sub-20ms real-time budget on edge hardware, 10% faster at 92% mAP
- Wrote 40+ C++ unit tests (80% coverage) for perception and set up Gazebo simulations, cutting track testing by 30%

SRKK Group **Kuala Lumpur, MY**
Summer Intern; Software Development (RAG) Aug 2025 – Sep 2025

- Built an Azure Power Automate RAG pipeline for 1000+ page financial RFPs, cutting human review time by 70%
- Lowered LLM hallucination by 30% through custom faithfulness scoring, grounding answers in retrieved chunks
- Reworked chunking with recursive character splitting, cutting query latency to 200ms and token use by 40%

EH Integrated Systems **Kuala Lumpur, MY**
Summer Intern; Full-Stack Java Software Engineer Jul 2024 – Sep 2024

- Cached invoice reads in Java, so checkout, refund and invoicing held ~100ms and <1% errors at 200K queries/day
- Containerized C# and Java back-end services with Docker for repeatable setups, cutting environment time by ~85%
- Designed and indexed a 12-table MySQL schema over 1M+ records, cutting inventory query latency to ~20ms

PROJECTS & OPEN SOURCE

Grady: UK Visa-Sponsor Graduate Jobs Pipeline Oct 2026 – Current

- Built a Python pipeline reading ~850 careers sites daily, fuzzy-matching employers to ~140k UK visa-sponsor records
- Made runs fail safe (retry backoff, per-site error isolation, no lost jobs); 198 pytest tests run in CI on push

Open-source contributor: Dify (langgenius/dify) Oct 2026 – Current

- Submitted 3 open PRs to Dify replacing mocks in 4 tests with real DB fixtures; added 1 tenant-isolation test
- Broke covered logic on purpose in all 3 PRs to prove tests fail; 209 surrounding tests and ruff pass

Automated methods for AI Video Detection *(Master's Thesis Project)* Sep 2025 – May 2026

- Achieved 97% precision, 0.9 AUC building a Video Swin Transformer for synthetic video detection (team of 5)
- Cut training epoch time by 35% on the shared AIRE HPC by staging data to flash, with I/O wait down 40%
- Led comparative analysis of 8 unique architectures; final ensemble model achieved highest recall (0.62)

CNN for Breast Cancer Classification *(Bachelor's Thesis Project)* Sep 2024 – May 2025

- Derived and coded backprop, ReLU activations, and weight updates by hand for a breast tumor CNN in NumPy
- Mitigated class imbalance using weighted loss functions and 5-fold CV, achieving 90.65% recall and 88.7% F1 score

EXTRACURRICULAR ACTIVITIES

DurHack Hackathon 2025 | Mario Kart Win Predictor **Durham, UK**
G Research Prediction Challenge, Honorable Mention Nov 2025

- Built a TensorFlow race predictor at 99% accuracy from 5000+ kart and map stats; top 2% of 100+ teams (team of 3)

British Chinese Society Treasurer **Leeds, UK**
Leeds University Society Sep 2023 – Jun 2024

- Managed a £2,800 budget for 200 members, allocating funds to raise £600 in sponsorship and hold 6 events on plan

ADDITIONAL INFORMATION

Languages:	English (Native), Mandarin (Native), Cantonese (Native), Malay (Intermediate), Japanese (Basic)
Technical Skills:	C++, C#, Java, C, Python, SQL, JavaScript, Docker, GitHub Actions, Azure, PyTorch, TensorFlow
Interests:	Grade 7 Piano, Grade 6 Violin, Leeds Badminton Club Team 2, Open-source Contributing